

Library Book Request Management System Using ServiceNow

PROJECT DOCUMENTATION

Objective

The objective of the Library Book Request Management System project is to automate and streamline the process of managing library book requests within an educational institution. The application enables students to submit book requests, allows librarians to review and update request statuses such as Pending, Approved, and Rejected, sends automated email notifications based on request status, provides request tracking capabilities, and supports efficient management of library services through a centralized and user-friendly platform.

Milestone 1: Environment Setup

Objective

To prepare the ServiceNow development environment and create the foundation for the Library Book Request Management System.

Step 1: Create Personal Developer Instance (PDI)

Visit developer.servicenow.com

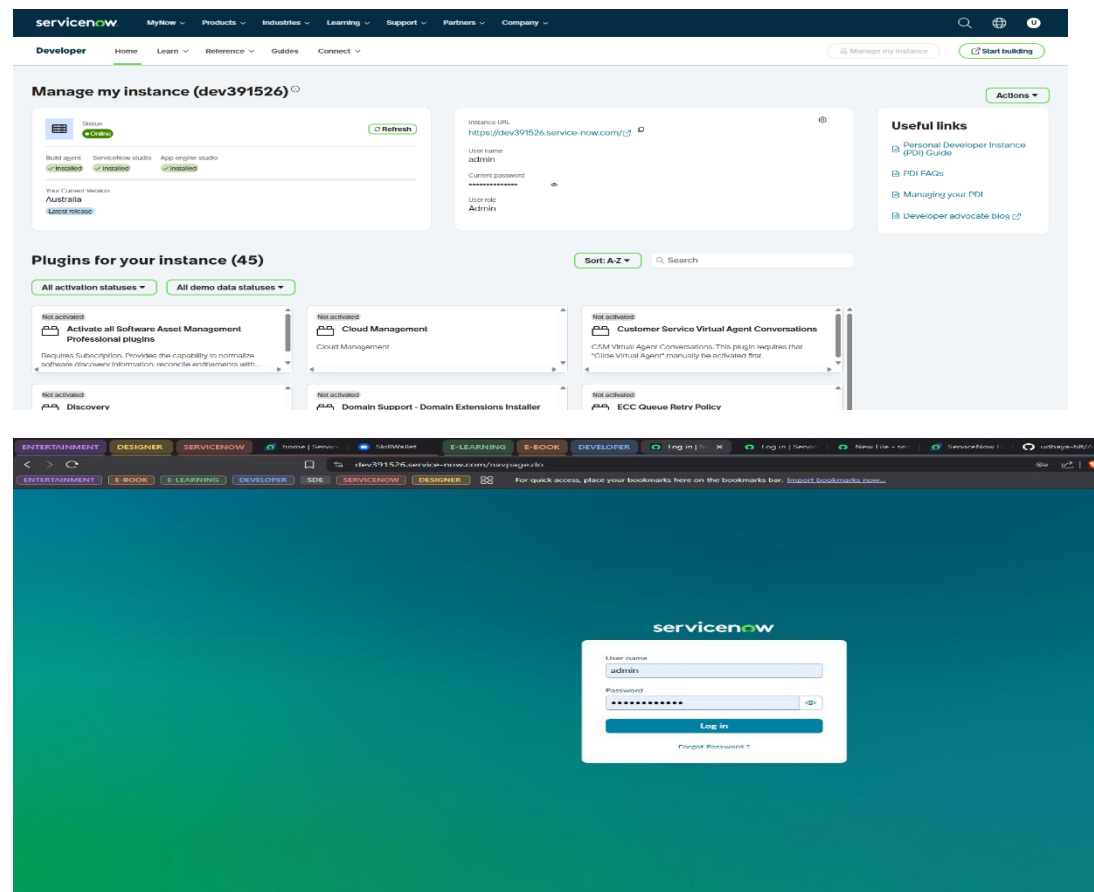
Sign in or create a developer account

Request a Personal Developer Instance

Wait for instance activation

Login using admin credentials

Screenshot



Step 2: Create Scoped Application

Navigation:

System Applications → Studio

Open Studio

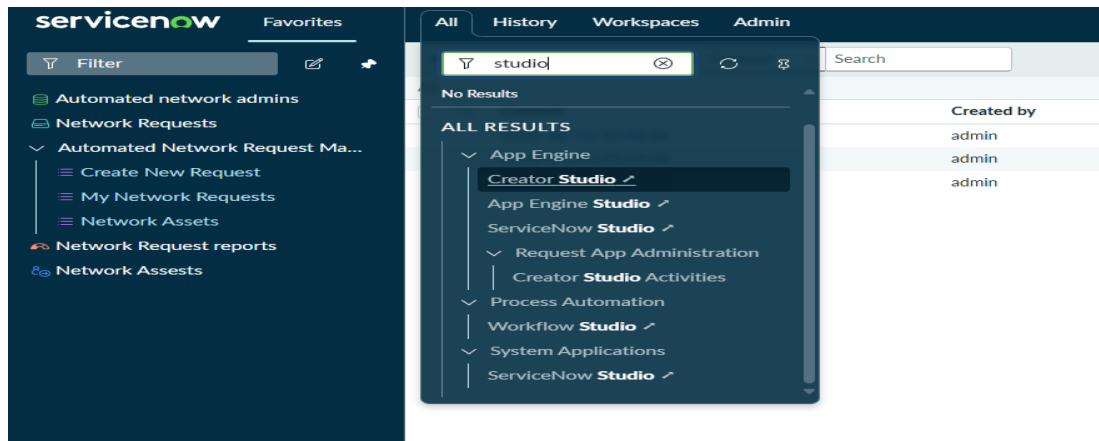
Select Create Application

Application Name:

Library Book Request Management System

Click Create

Screenshot



Learning Outcomes

- Understood ServiceNow architecture
- Created a scoped application
- Configured development environment

Milestone 2: Data Model Design

Objective

Design custom tables and fields for managing library book requests.

Step 1: Create Library Request Table

Navigation:

Studio → Create Application File → Table

Table Name:

Library Request

Fields:

- Request Number
- Student Name
- Student Email
- Book Name
- Issue Date
- Status

Screenshot

library request

Data

Forms

Flows

Policies and rules

Library Book R...

Preview

Save

Table fields

+ Add new field

More actions

Filter fields

Filter options

Column label *	Column name *	Type * 17	Reference	Max length	Default value	Display	Updated
approval	approval	String		40			2026-06-16 08:14:02
Student id	student_id	String		40			2026-06-14 21:41:58
Book name	book_name	String		40			2026-06-14 21:41:58
Student name	student_name	String		40			2026-06-14 21:41:59
Number	number	String		40	javascript:global.getNextO...		2026-06-14 21:50:21
Updated by	sys_updated_by	String		40			2026-06-14 21:34:57
Request number	request_number	String		40			2026-06-14 21:41:59
student email	student_email	String		40			2026-06-16 08:14:01
Created by	sys_created_by	String		40			2026-06-14 21:34:57
Updates	sys_mod_count	Integer					2026-06-14 21:34:57
Updated	sys_updated_on	Date/Time					2026-06-14 21:34:57
Created	sys_created_on	Date/Time					2026-06-14 21:34:57
Issue Date	issue_date	Date					2026-06-14 21:41:58
approval status	approval_status	Choice 4 Choices					2026-06-15 21:27:00
status	status	Choice 4 Choices			pending		2026-06-17 08:17:27

Add new field

Learning Outcomes

- Created custom tables
- Designed application data model
- Configured fields and forms
- Built navigation modules

Milestone 3: Business Logic and Validation

Objective

Implement automation and validation for request processing.

Step 1: Configure Form Layout

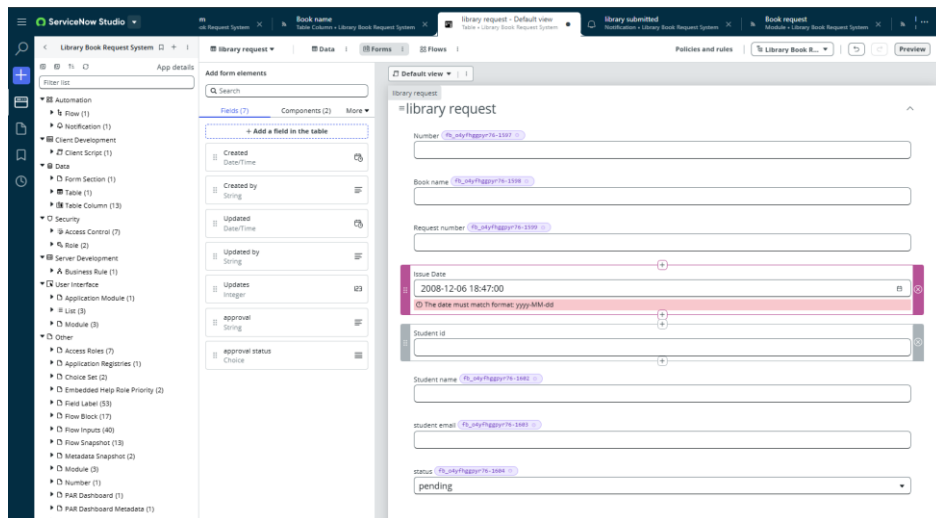
Navigation:

Open Library Request Form

Right Click → Configure → Form Layout

Arrange fields logically.

Screenshot



Step 2: Create Client Script for Issue Date Validation

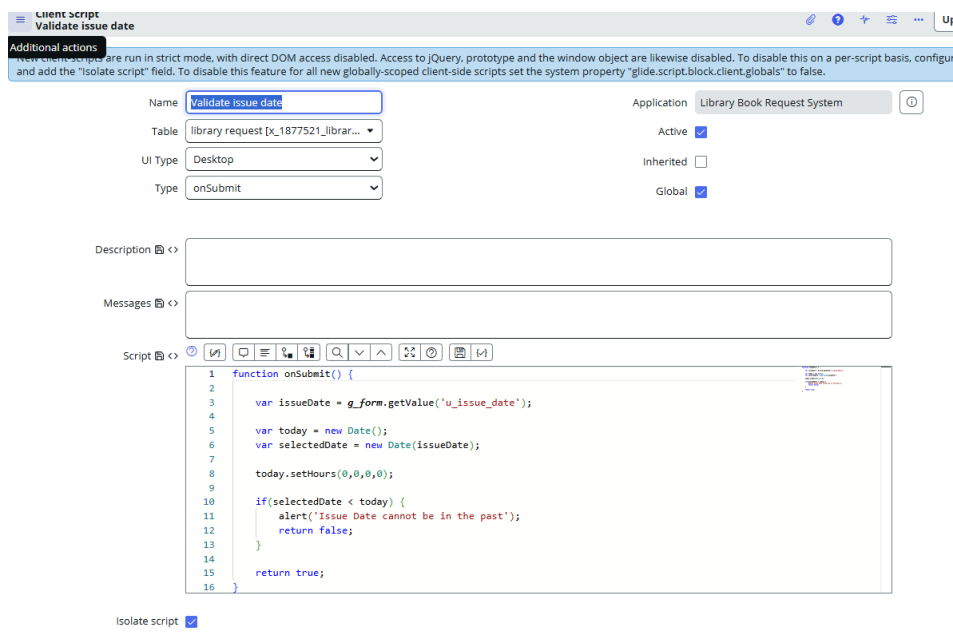
Purpose:

Prevent users from selecting past dates.

Validation:

Issue Date cannot be earlier than the current date.

Screenshot :



Step 3: Auto Generate Request Number

Purpose:

Automatically generate unique request numbers.

Examples:

LBR01001

LBR01002

LBR01003

Learning Outcomes

- Implemented client-side validation
- Automated request numbering
- Improved data accuracy
- Enhanced user experience

Milestone 4: Workflow Automation and Notifications

Objective

Automate request processing and communication using Flow Designer.

Step 1: Create Status Management Flow

Purpose:

Automate student notifications based on request status.

Statuses:

- Pending
- Approved
- Rejected

Workflow:

Request Created or Updated

↓

Check Status

↓

Pending / Approved / Rejected

↓

Send Appropriate Email Notification

Screenshot:

Application scope: Library Book Request System
Update set: Default [Library Book Request System]

Emails View: Outbox

Created	Recipients	Subject	State
2026-06-17 00:00:13	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	Ready
2026-06-17 01:00:03	allen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	Ready
2026-06-17 01:00:03	allen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	Ready
2026-06-17 01:00:04	allen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	Ready
2026-06-17 08:43:24	librarian@test@gmail.com	New library book request	Ready
2026-06-17 08:46:31	pradeep@gmail.com	library request was rejected	Ready
2026-06-17 08:46:31	pradeep@gmail.com	library request approved	Ready
2026-06-17 08:47:07	pradeep@gmail.com	library book request submitted successfully	Ready
2026-06-17 09:18:02	librarian@test@gmail.com	New library book request	Ready
2026-06-17 09:18:57	ritish@gmail.com	library request approved	Ready
2026-06-17 09:24:35	krish@gmail.com	library request approved	Ready
2026-06-17 09:25:11	krish@gmail.com	library book request submitted successfully	Ready
2026-06-17 09:38:37	ronco@gmail.com	library request approved	Ready
2026-06-17 09:49:42	messi@gmail.com	library book request submitted successfully	Ready
2026-06-17 09:50:39	messi@gmail.com	library request approved	Ready
2026-06-17 10:01:03	admin@example.com	[TSM Analytics] Daily Data Collection completed with error	Ready

library request update

TRIGGER

library request Created or Updated

ACTIONS

1. If approved

2. Send Email

3. Else if rejected

4. Send Email

5. Else if requested

6. Send Email

ERROR HANDLER

If an error occurs in your flow, the actions you add here will run.

Data

Flow Variables

Trigger - Record Created or Updated

library request Record

Changed Fields

library request Table

Run Start Time UTC

Run Start DateTime

1. If

2. Send Email

email

Action Status

3. Else If

4. Send Email

email

Action Status

5. Else If

6. Send Email

email

Action Status

Step 3: Verify Email Notifications

Navigation:

System Mailboxes → Outbox

Verify generated emails.

Screenshot

Library book pending:

Workspaces Admin

Email - library book request submitted successfully

Update set: Default [Library Book Request System]

request submitted successfully View: Outbox

Mailbox: Outbox

State: Ready

Target: (empty)

User:

Subject: library book

Recipients: vishal@gmail.com

Body:

Preview Email

Dear vishal

Your library book request has been submitted successfully and is currently under review.

Request Details:

- Request Number 647383
- * Student Name: vishal

* Book Name : harry potter

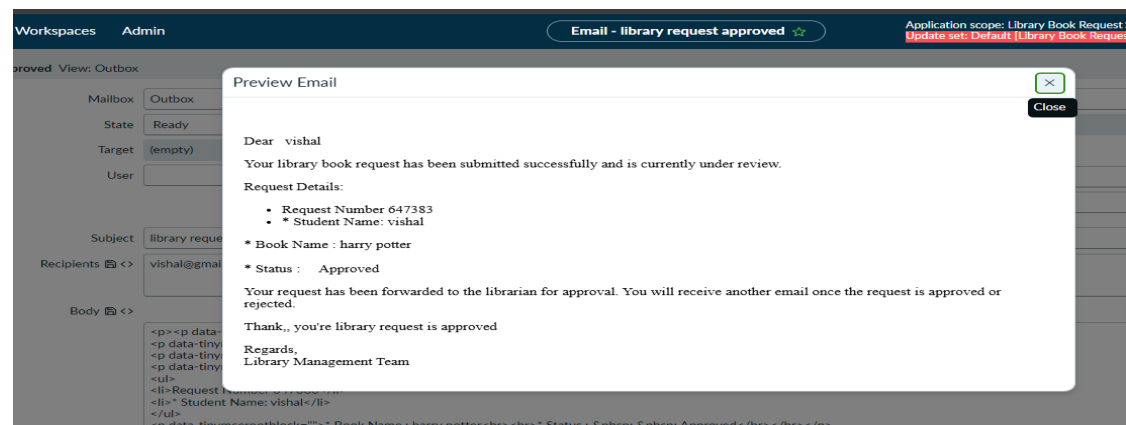
* Status : pending

Your request has been forwarded to the librarian for approval. You will receive another email once the request is approved or rejected.

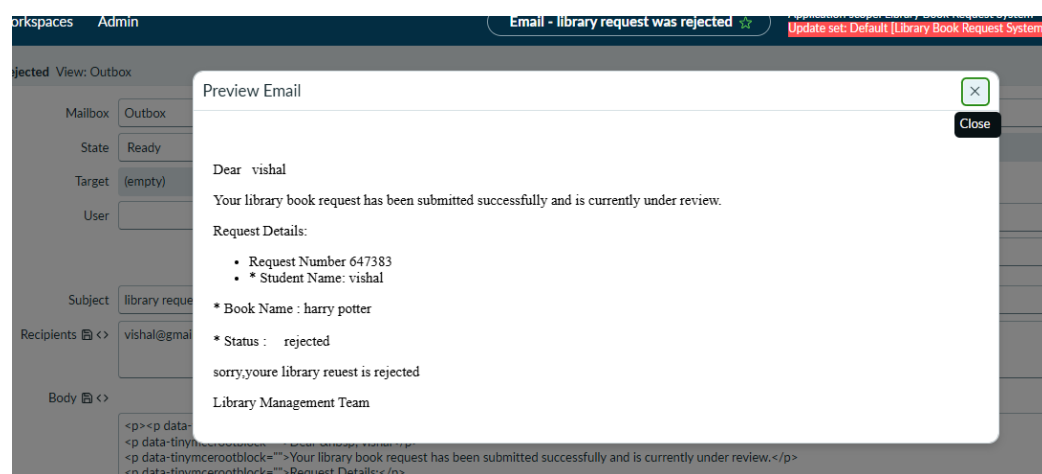
Thank you for using the Library Book Request System.

Regards,
Library Management Team

Library book approved:



Library book rejected:



Learning Outcomes

- Created Flow Designer workflows
- Automated email notifications
- Implemented status-based automation
- Improved communication process

Milestone 5: Request Management and Tracking

Objective

Enable efficient tracking and management of library requests.

Features Implemented

- Request Submission Form

- Request Status Tracking
- Pending Status Handling
- Approval Processing
- Rejection Processing
- Email Notifications

Screenshot:

Book name	Issue Date	Number	Request number	Student id	Student name	status
Search	Search	Search	Search	Search	Search	Search
(empty)		LBRO1002				
java programming	2026-06-15	LBRO1004	1233444	717823y134	nithish	rejected
c	2026-06-14	LBRO1005	232424	23w32233	nithrt	rejected
java	2026-06-16	LBRO1006	123756	56556	nithish	rejected
react	2026-06-10	LBRO1010	3456	7hyvy778	vushal	Approved
anime	2026-06-17	LBRO1012	123545	1323e22332	ganesh	Approved
vgvuy	2026-06-17	LBRO1014	1234556	5656f	vibuh	Approved
grgr	2026-06-17	LBRO1015	2312323	343vfree	dfd vfd	Approved
vishia	2026-06-17	LBRO1016	256	7565765	yuhgyuygyu	rejected
jungle book	2026-06-18	LBRO1017	34565	1234e888	vishalrupa	Approved
tamil book	2026-06-27	LBRO1020	2123444	1213243r3434	pradeep kumar	pending
Its ends with us	2026-06-22	LBRO1022	455445	37981738991	nithish	Approved
Life along with me	2026-06-23	LBRO1023	23332	1213242	krish	pending
one day star	2026-06-22	LBRO1025	1515	121213w	ronaldo	Approved
python	2026-06-21	LBRO1026	12131	1555545	messi	Approved

Approved

Book name	Issue Date	Number	Request number	Student id	Student name	status
Search	Search	Search	Search	Search	Search	Search
(empty)		LBRO1002				
java programming	2026-06-15	LBRO1004	1233444	717823y134	nithish	rejected
c	2026-06-14	LBRO1005	232424	23w32233	nithrt	rejected
java	2026-06-16	LBRO1006	123756	56556	nithish	rejected
react	2026-06-10	LBRO1010	3456	7hyvy778	vushal	Approved
anime	2026-06-17	LBRO1012	123545	1323e22332	ganesh	Approved
vgvuy	2026-06-17	LBRO1014	1234556	5656f	vibuh	Approved
grgr	2026-06-17	LBRO1015	2312323	343vfree	dfd vfd	Approved
vishia	2026-06-17	LBRO1016	256	7565765	yuhgyuygyu	rejected
jungle book	2026-06-18	LBRO1017	34565	1234e888	vishalrupa	Approved
tamil book	2026-06-27	LBRO1020	2123444	1213243r3434	pradeep kumar	pending
Its ends with us	2026-06-22	LBRO1022	455445	37981738991	nithish	Approved
Life along with me	2026-06-23	LBRO1023	23332	1213242	krish	pending
one day star	2026-06-22	LBRO1025	1515	121213w	ronaldo	Approved
python	2026-06-21	LBRO1026	12131	1555545	messi	rejected

rejected

Emails View: OutlookCreated - Search

Actions on selected rows...

All > Mailbox > Outlook > Created on Today

Created

Recipients

Subject

State

Search	about a minute ago	Search	Search
☐	☐	☐	☐
2026-06-17 23:38:26	messi@gmail.com	library request was rejected	Ready
2026-06-17 10:01:03	admin@example.com	[ITSM Analytics] Daily Data Collection completed with error	Ready
2026-06-17 09:50:39	messi@gmail.com	library request approved	Ready
2026-06-17 09:49:42	messi@gmail.com	library book request submitted successfully	Ready
2026-06-17 09:38:37	rono@gmail.com	library request approved	Ready
2026-06-17 09:25:11	krish@gmail.com	library book request submitted successfully	Ready
2026-06-17 09:24:35	krish@gmail.com	library request approved	Ready
2026-06-17 09:18:57	nithin@gmail.com	library request approved	Ready
2026-06-17 09:18:02	librarintest@gmail.com	New library book request	Ready
2026-06-17 08:47:07	pradee@gmail.com	library book request submitted successfully	Ready
2026-06-17 08:46:31	pradee@gmail.com	library request approved	Ready
2026-06-17 08:45:24	pradee@gmail.com	library request was rejected	Ready
2026-06-17 08:43:24	librarintest@gmail.com	New library book request	Ready
2026-06-17 01:00:04	aleen.motern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	Ready
2026-06-17 01:00:03	aleen.motern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	Ready
2026-06-17 01:00:03	aleen.motern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	Ready
2026-06-17 00:02:13	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	Ready

Learning Outcomes

- Managed request lifecycle
 - Implemented workflow tracking
 - Improved operational visibility
-

Testing

Test Case 1

Student submits a book request.

Expected Result:

Status becomes Pending.

Result:

Passed.

Test Case 2

Librarian approves request.

Expected Result:

Status changes to Approved.

Result:

Passed.

Test Case 3

Librarian rejects request.

Expected Result:

Status changes to Rejected. Result:

Passed.

Test Case 4

Email notification generated.

Expected Result:

Email sent successfully.

Result:

Passed.

Milestone 6: Dashboard and Reporting

Objective

Provide visual insights into library book requests and enable librarians to monitor request trends through reports and dashboards.

Step 1: Configure Dashboard

Self-Service → Dashboards

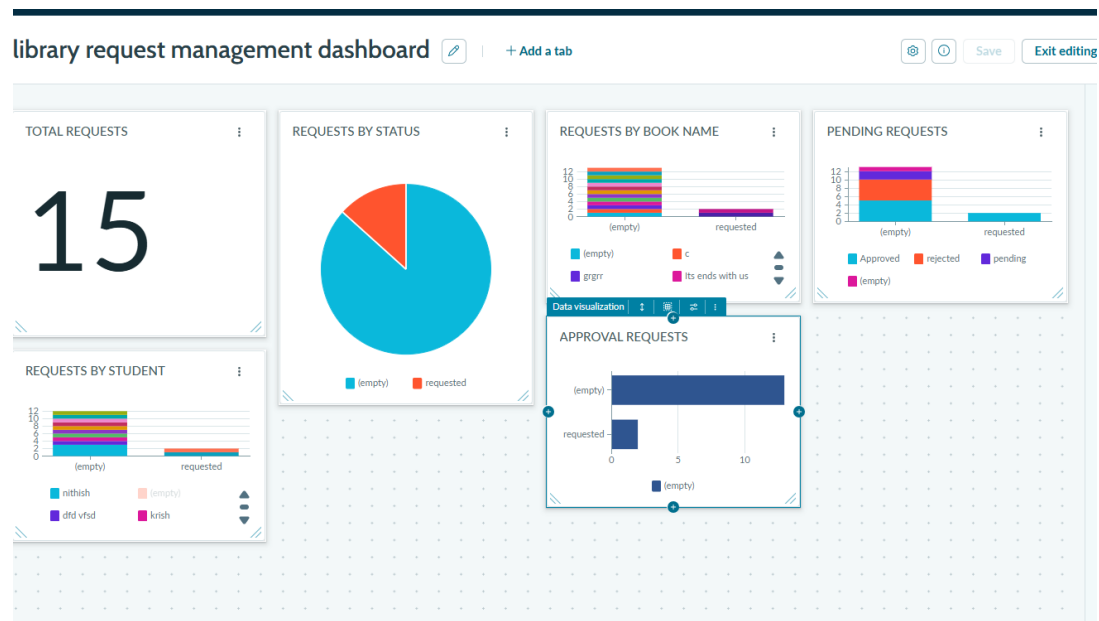
Dashboard Name:

Library Request Management Dashboard

Components Added:

- Total Requests
- Pending Requests
- Approved Requests
- Rejected Requests
- Request Status Distribution Chart

Screenshot



Achievements:

- Online request submission.
- Automated approval workflow.
- Real-time status tracking.
- Email notifications.
- Centralized request management.

Conclusion

The Library Book Request Management System successfully automates the lifecycle of library book requests using the ServiceNow platform. The project demonstrates key ServiceNow concepts including scoped application development, custom tables, client scripts, Flow Designer automation, notifications, and request tracking. The solution improves efficiency, reduces manual effort, and provides better visibility into the library request management process.

Future Enhancements

- Library Dashboard and Reporting
- Book Availability Tracking
- Return Date Management
- Multi-Level Approval Process
- Service Portal Integration
- Mobile Accessibility
- Advanced Analytics Dashboard
- AI-Based Book Recommendations
- Integration with Library Inventory Systems